

Response to Examiner Questions

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Thank you for reading the thesis so closely and for putting all of these questions to me. Several of them sent me back to my own results, and two of them turned up mistakes I had not spotted. This was my first major research project, and I know I still have a lot to learn about research, analysis, and presenting my work. I genuinely tried and gave my best throughout, but I also recognise that I made mistakes along the way. I have also answered each question on its own below, and I have tried to be straight throughout, including where the honest answer is that I got something wrong or did not test something I should have. I accept these mistakes and see them as part of my learning, and I hope to learn from this experience and do better in my future work.

Question 1

Of the handful of results that did come back significant, such as the hardness accuracy gap, the dissolution improvement from sharing one encoder, and the attribution shift on dissolution and hardness: how confident can we be that these are real, given how many tests were run without any correction for that?

Answer 1. You are right. I should have corrected for this and I did not. It was an oversight rather than a decision I made and defended, and the report says nothing about Benjamini–Hochberg or Bonferroni anywhere.

I ran twelve tests: four in RQ1 against tuned gradient boosting, four in RQ2 for multi-task against single-task, and four in RQ3b for the attribution rankings, one per target. Here is what happens when I correct across all twelve.

Test	Raw p	Benjamini–Hochberg	Bonferroni
Attribution, dissolution	0.0004	0.0024 survives	0.0048 survives
Attribution, hardness	0.0004	0.0024 survives	0.0048 survives
Multi-task gain, dissolution	0.012	0.045 survives	0.144 fails
Hardness gap vs. gradient boosting	0.015	0.045 survives	0.180 fails
Attribution, weight RSD and tensile	0.073	0.145 fails	0.871 fails

Taking your three in turn.

The attribution p -value comes through both corrections without trouble. I want to be careful here, though, because that number is easy to over-read. All the test tells you is that the two rankings, confident and unconfident, are related rather than independent of each other. The bigger claim sitting on top of it, that the model leans much harder on certain channels when unsure, is a different claim, and it never got a test of its own. So the p -value that survived is carrying less weight than the report suggested. Question 8 is where I deal with that properly.

The dissolution gain and the hardness gap both land on 0.045 under Benjamini–Hochberg. That is inside the line, but only just, and both fail Bonferroni outright. They are borderline results. The report should have said so instead of calling them significant and leaving it there.

A note on the count, because it affects the sums. Two pairs of targets in RQ3b return identical statistics: dissolution matches hardness, and weight RSD matches tensile. I wondered about that, so I checked rather than assumed. It falls out of the sample size. With ten channels, τ depends only on how many discordant pairs there are, which leaves 46 possible values in total. A τ of 0.822 is exactly 4 discordant pairs and 0.467 is exactly 12, so two targets landing together is not surprising at all.

It also does not matter for the answer. Count those pairs as two tests instead of four, so ten rather than twelve, and the same findings survive each correction.

There is one thing in my favour, though it does not excuse the omission. The report already treats every non-significant result as inconclusive rather than as proof the models are equal, because three folds leave the test with very little power. So the correction makes me more cautious. It does not overturn anything.

Question 2

Your work treats product code 15 as the key test of how the model handles a genuine distribution shift (Section 3.5, Section 6.3) and finds the error jumps to 2.43 times the baseline on it. But code 15 was also the code entirely excluded from training in the first place, specifically because it was too large and would dominate a fold (Section 3.4). So when the model does badly on code 15, it is not clear whether that is because the model genuinely struggles with an unfamiliar hardness regime, or simply because it never saw a single example anything like it during training, which I think is a much less interesting and less surprising finding. How would this “distribution shift” result look different from ordinary “the model has literally never seen this” behaviour, and is there a way to tell those two explanations apart?

Answer 2. This is the exact confound I was worried about when I designed the experiment, and code 25 is there to separate the two.

Neither code appears anywhere in the fold 1 model’s training data. Both are equally new to it. The one thing that differs is where their hardness sits.

	Code 25	Code 15
Seen anywhere in training	No	No
Batches	34	64
Mean hardness	50.4 N	90.2 N
Hardness RMSE	4.918 N	18.69 N
Against CV reference of 7.698 N	0.64×	2.43×

That table is the answer to your question. Code 25 is just as new to the model as code 15 is. If never having seen something were the problem, code 25 should suffer too. It does not. Its error sits about a third below the cross-validation baseline, so being an unfamiliar product costs nothing on its own. The novelty is the same for both. Only the hardness regime differs, and only that produces a penalty.

I can do better than one pair, though. Grouping the folds by product code means all 24 codes inside cross-validation are already leave-one-code-out results, free of charge, and every one was scored by a model that had never trained on it. If novelty were driving this, all 24 should suffer. Sorted by distance from the centre of the training hardness distribution, they do not:

Distance from training centre	Codes	Mean hardness	RMSE
Within 3 N	12		4.55 N
Between 3 N and 10 N	5		5.38 N
More than 10 N	7		24.65 N

Across all 24, error against distance from the centre gives $r = 0.941$ with $p = 7.7 \times 10^{-12}$. That is Figure 3b in the report.

Some of those codes are small, so I went back and checked whether they are propping the correlation up. Seven of the 24 have five batches or fewer, and between them they hold 18 batches out of 941. Weight each code by its batch count and r comes to 0.856. Keep only the 10 codes with at least 20 batches and it comes to 0.947. So a handful of tiny codes is not what is producing this. I should add that the relationship is not a straight line either: error climbs far faster at the far end than near the centre, which is what those three bands show.

Every one of the 24 is equally novel. Novelty cannot produce a correlation that strong. Distance from the trained range can.

One concession. Code 15 sits right at the end of that line and it was held out for a structural reason in the first place, so taken on its own it is not a clean test. You are right about that. The 24-code analysis is the better evidence, and Section 6.3 should have led with it instead of putting code 15 forward as the headline.

Question 3

Why did one weak result (hardness accuracy) get a full root-cause investigation while the other one (weight uniformity's uncertainty calibration, roughly five times worse than the other three targets) was just noted and left as a limitation? Was there an attempt to understand why weight uniformity's intervals don't respond to the requested confidence level, or was that simply out of scope?

Answer 3. It was out of scope, and if I am honest that came down to time rather than any real decision. Hardness sits in the main accuracy table, so it is the first weakness a reader runs into, and that is where I put the effort. The weight uniformity problem is one line in a table. I flagged it and moved on. Nobody looked into the cause before I submitted.

I have looked since, and it behaves exactly as you say. Its coverage goes from 0.930 at the 80 per cent level to 0.960 at 95 per cent. That is a spread of 0.030, against 0.127 for dissolution, 0.111 for hardness and 0.075 for tensile. It starts over-covered and then hardly moves.

I have a hypothesis about the cause, which I want to label as a hypothesis before I give it, because I have not tested it. It concerns the shape of the target. Weight RSD is the only heavily skewed one of the four. Its median is 0.97, its 95th percentile is 1.86, and its maximum is 9.90, so the maximum is more than ten times the median. The other three targets have maximum-to-median ratios between 1.1 and 1.9.

The evidential head puts a Normal-Inverse-Gamma distribution on the output, and that is symmetric. Fit a symmetric interval to a skewed target and one scale factor has to serve two very different regions at once, the crowded centre and the long tail. Whatever scale is wide enough for the tail will be far too wide for everything else. That is what a flat coverage curve looks like.

To be clear about what that paragraph is: a hypothesis that fits the evidence, and nothing more. It is not a finding, and I did not test it. The experiment that would confirm or refute it is to model log weight RSD instead and check whether the coverage curve straightens out. I have not run that experiment, so I cannot tell you whether the explanation is right.

What I am sure about is the part the report already states. Do not rely on those intervals for a single batch. The uncertainty–error correlation for weight RSD is +0.029, which is effectively nothing, so the interval tells you nothing about which batch is going to be wrong.

Question 4

You evaluate your model against a deep ensemble baseline to assess the quality of its uncertainty estimates. However, unlike the rest of the study, this comparison is restricted to a single fold rather than the full three-fold cross-validation. While you acknowledge that this provides only a preliminary indication, the conclusion still asserts that the proposed model captures “most of the benefit” of the ensemble at significantly lower computational cost.

Answer 4. That is a real inconsistency in the report and I accept it.

Section 6.3 says the single-pass evidential model “provides most of the benefit at roughly one-fifth of the training and inference cost”. Two claims are doing work in that sentence, and they do not rest on the same footing.

The cost claim is just arithmetic, so it holds no matter how many folds were run. A five-seed ensemble is five training runs and five forward passes against one, so one-fifth is a count, not a measurement.

The benefit claim is a different matter. “Most of the benefit” is quantitative, and a single fold cannot support a quantitative claim. Labelling the comparison as directional in the following sentence does not repair it, because the strong claim has already been made. I should have written that the comparison is preliminary and indicates a direction, without attaching a size to it.

The underlying numbers are also more mixed than that phrase suggests. Under distribution shift on code 15, the ensemble is more accurate on all four targets:

Target	Ensemble	Evidential	Ensemble better by
Dissolution	2.793	2.895	3.5%
Hardness	14.08	18.69	24.7%
Weight RSD	0.342	0.364	6.0%
Tensile	0.117	0.143	18.2%

On the same fold in distribution, though, two of those flip round. My model gives 3.344 on dissolution where the ensemble gives 3.408, and 7.900 on hardness where the ensemble gives 9.081.

So here is the version I would stand behind. On one fold, the ensemble holds up better under distribution shift on all four targets, and clearly better on hardness. The cost ratio is fine as it stands. But “most of the benefit” attaches a size to something I measured once, and that should not have gone in.

Question 5

Given the single-fold evaluation, how robust is this conclusion? Why was the full three-fold evaluation not conducted for this baseline to ensure parity with the primary results?

Answer 5. There are two parts to this, so let me take them in turn.

How robust it is. Not robust enough for the way the report phrased it. One observation per target means no spread to look at and nothing to test. The direction is at least consistent, in that the ensemble wins on all four targets under shift on code 15 rather than splitting them. But those four are not four independent observations. Same model, same held-out data, same seed.

Table 4 shows why one fold is especially weak here. MT-TrajNet varies by $1.06\times$ between folds on dissolution and hardness, but $1.74\times$ on weight RSD and $1.43\times$ on tensile. On those two noisier targets, the fold-to-fold swing is about the same size as the model differences I am reporting. One fold cannot separate the two.

The hardness comparison is the piece I would defend: 14.08 N against 18.69 N, a gap of 24.7 per cent. That is bigger than MT-TrajNet’s own $1.06\times$ hardness swing across folds, so it is probably not just noise. The other three gaps at 3.5, 6.0 and 18.2 per cent I would not defend on a single fold.

Why three folds were not run. Compute. A five-seed ensemble means five full training runs per fold, so three folds would have required fifteen runs against the three the main model needed. The work ran on the free Kaggle tier with a twelve-hour session limit and a single Tesla T4. Fifteen runs alongside the rest of the study was more than I could complete reliably, so I ran fold 1 and labelled the result as directional.

Two things make that explanation less comfortable than it sounds, and I should put both on the record.

The five seeds came straight from my proposal, which specified a five-seed ensemble. So the number was carried over rather than chosen for this particular experiment.

And fifteen runs was never the only option. Three seeds across three folds is nine runs, fewer than fifteen, and would have bought me fold coverage in exchange for a smaller ensemble. I did not weigh that up at the time. There is no note about it anywhere in my notebooks. Looking back, fold coverage was probably the better use of the compute, because fold-to-fold variation is exactly the thing a single fold cannot tell you about.

Leaving it at one fold was still a fair call given what I had to work with. Letting that single fold carry a conclusion phrased that strongly was not. Given the same compute again, I would either run all three folds and drop something else, or keep the one fold and report only the direction.

Question 6

The hardness root-cause analysis in Section 6.5 explains the model’s weak hardness accuracy by pointing to how few high-hardness batches remain once code 15 is withheld. But that same scarce, code-concentrated hardness data is also what produces your strongest positive result, the attribution shift with tau equal to 0.822 on hardness in Section 6.4. Is it possible that the same structural quirk, a handful of codes driving both the high-hardness examples and the uncertainty signal, is inflating the attribution finding, rather than that finding reflecting genuine learning of physically relevant sensors?

Answer 6. This is the hardest of the ten, and when I went and looked, part of what you are pointing at turns out to be true.

Fold 1 has 314 test batches. Only one high-hardness code sits in it, code 4, with 22 batches. Split those 314 into quintiles by hardness uncertainty and every one of the 22 code 4 batches lands in the least-confident fifth, with none in the most-confident fifth. So code 4 is 35 per cent of the low-confidence group and nothing at all of the high-confidence group. The concentration you describe is real and it is sitting in my data. I had not checked it before submitting and I should have.

That is not the end of it, though, because there is a test that separates the two explanations. If code 4 were doing all the work, taking it out should kill the finding.

Excluding code 4 and recomputing the quintiles on the remaining 292 batches, both groups become dominated by the same product code, code 13, at 86 per cent of the low-confidence group and 81 per cent of the high-confidence group. Their mean true hardness becomes 50.95 N and 51.14 N, essentially identical. Yet the model still separates them by a factor of 3.34 in its own uncertainty, 41.14 against 12.30.

The cleanest version of the test stays inside one product code. Code 13 has 131 batches, all the same product, all in the low-hardness range:

Within code 13 only	Least confident 20%	Most confident 20%
Mean uncertainty	51.40	10.69
Mean true hardness	50.4 N	52.8 N
Mean absolute error	9.12 N	4.38 N

Same product, same hardness range, and the model still separates confident from unconfident batches by 4.8 times in uncertainty. More importantly, that separation tracks reality: the batches it is less sure about are the ones it actually gets more wrong, by a factor of 2.08.

So the uncertainty is not simply a restatement of which code a batch came from. It works inside one product code, at one hardness level, and it still tells you something about where the error is.

There is a limit to how far I can push this and I would rather say so than let it pass. I cannot rerun the SHAP attribution with code 4 taken out, because I saved the aggregated per-channel ratios instead of the per-batch SHAP values. So I can show the uncertainty signal itself survives the confound. I cannot show directly that the 4.86 and 2.81 ratios survive it. Rerunning that analysis without code 4 is the test that would settle it, and it is the first thing I would do with more time.

Question 7

Your calibration correction is cross-fitted so that no batch is calibrated on its own prediction (Section 3.6), and you note that an earlier, leakier version produced almost the same coverage numbers. Does that similarity not raise a concern about how sensitive this calibration step actually is? If the correction were doing meaningful work, would you not expect the numbers to move more once the leak was fixed? What would tell you the calibration step is not effectively a no-op given your sample size?

Answer 7. Two different things are being run together here, and the report is what mixed

them, which is why the question comes up at all.

The calibration step itself is doing a great deal of work. The fitted scale factors are 0.212 for dissolution, 0.317 for hardness, 0.570 for weight RSD and 0.255 for tensile. Because the raw predictive standard deviation is multiplied by that scale, the raw evidential intervals are between 1.75 and 4.72 times wider than they should be, depending on the target. Without the correction, coverage would sit near 1.0 on every target, which is useless for deciding anything. So the step is very far from a no-op.

What barely moved is something much narrower: the difference between fitting that scale on the evaluation data and cross-fitting it on the other folds. Your reasoning about why is correct, and the explanation is that the scale factors are extremely stable across folds. For hardness they are 0.327, 0.311 and 0.312, varying by about 5 per cent. When the quantity being estimated is that stable, it makes very little difference which subset it is estimated from, so removing the leak changes almost nothing.

That stability is really the answer to your first question. One multiplicative scalar per target, fitted on several hundred batches, is about as low-variance a thing as you can estimate. It is nothing like fitting a model with many parameters, where a leak can inflate the result badly. So a small change is exactly what you would expect here. It is not a warning sign.

Fixing it was still right. Reporting a number fitted on its own evaluation data is not sound whether or not it moves the answer. What I should not have done is present the small change as though it were reassuring or surprising. It follows straight from the scale being stable, and the report should have said that rather than leaving the observation hanging.

To your last question directly. Two things would tell me the step was ineffective. First, scale factors close to 1, which would mean the raw intervals were already about the right width and the correction was doing nothing. Second, scale factors varying widely between folds, which would mean no single value was meaningful and the whole approach was unstable. Neither is the case here. The scales are far from 1 and they are consistent across folds, which is the combination that says the step is both necessary and reliably estimated.

Question 8

You use Kendall's tau to compare SHAP rankings between the most confident and least confident batches (Section 3.7 and 6.4), treating $p < 0.05$ as evidence that the model shifts which sensors it relies on. But this comparison involves ranking only ten channels. How much statistical power does Kendall's tau really have at that scale, and could a similarly significant tau appear by chance if you randomly permuted which batches fell into the confident and unconfident groups?

Answer 8. I will take the power question first, and then raise something bigger that your question led me to find.

On power. The exact two-sided p -value for $\tau = 0.822$ at $n = 10$ is 0.000358, which is what the report rounds to 0.0004. I confirmed this with the exact method in `scipy` and with a permutation check over 200,000 random ranking pairs, which puts the one-sided probability at roughly 0.0002. For contrast, $\tau = 0.467$ has an exact p of 0.0726, which is why the weight RSD and tensile result falls outside significance. So the p -values printed in the report are arithmetically correct for the test that was run.

On your permutation question, which is a different test, and which I did not run. I want to be clear about what I tested and what I did not.

What was tested: the permutation of the two rankings, described above. That is the check behind the 0.0002 figure.

What was not tested: the permutation you actually describe, shuffling which batches land in the confident and unconfident groups. Those are different null hypotheses. Yours is the harder one, because it asks whether splitting by uncertainty achieves anything a random split of the same batches would not.

I did not run that test and I cannot run it now. Notebook 11 saved the aggregated per-channel ratios instead of the per-batch SHAP values, so the inputs I would need to re-split the batches and recompute the rankings are gone short of running the whole analysis again. That is down to how I saved my results. I would rather admit it than hand you the test I can run and let it pass for the one you asked for.

I have no result to report for it, and I am not going to imply one. What I can give you is a prediction, and it is reasoning rather than evidence. Two random halves of the same batches would both be estimating the same underlying attribution, so their rankings should agree closely and τ should sit near 1. If that reasoning holds, a random split would give a higher τ than the 0.822 observed for the confidence-based split. Whether it does hold is unknown to me, because the measurement was never made. Re-running notebook 11 with the per-batch SHAP values retained, then permuting the group labels a few thousand times, is the experiment that would settle it, and it is the first thing I would run given more time.

Now the more important point. While checking this I found that the report interprets τ backwards, and I would rather raise it myself than leave it.

Section 6.4 states: “The rankings shift significantly on dissolution and hardness ($\tau = 0.822$, $p = 0.0004$).” But Kendall’s τ measures whether two rankings agree. A τ of 0.822 means they are strongly concordant, only four discordant pairs apart out of forty-five. The p -value tests the null hypothesis that $\tau = 0$, meaning no relationship at all. So $p = 0.0004$ says the confident and unconfident rankings are significantly correlated, which is evidence that the sensor ordering is stable, not that it changed.

The statistic I cited does not test the claim I made with it. That is an error in the report and I accept it.

The underlying finding is still real, but it is carried by different numbers. What actually changes when the model becomes uncertain about hardness is the magnitude of its reliance on each channel:

Channel	Ratio	Channel	Ratio
die fill	$4.86\times$	tbl speed	$1.38\times$
cyl main	$2.81\times$	SREL	$1.28\times$
main comp	$1.58\times$	pre comp	$1.15\times$
cyl pre	$1.52\times$	ejection	$1.01\times$
fom	$1.46\times$	stiffness	$0.98\times$

Read together with $\tau = 0.822$, the correct interpretation is that the model does not reorder which sensors it uses. It keeps roughly the same ordering and leans far harder on two specific channels, die fill and main compression, when it is uncertain. Ejection and stiffness barely move at all.

Before I say what those ratios mean, I need to be straight about what they are. They are

descriptive numbers, nothing more. I never attached a significance test to them, and there is no confidence interval or standard error saved with them. The only inferential statistic in this whole analysis is τ , and τ tests rank agreement, not the size of a ratio. So the amplification pattern is something I observed in the data, not a result I established statistically. The report should not have called it the strongest statistical result in the study. That label came from τ 's p -value, and I now accept the p -value was attached to the wrong claim.

One piece of supporting context that the results file does contain, and that I did not use in the report. The mean epistemic uncertainty in the two groups is 6.688 for the confident fifth and 54.654 for the unconfident fifth, a separation of about 8.2 times. That is not a significance test either, but it does show the two groups are genuinely far apart on the variable used to split them, so the comparison is not being drawn between two groups that are effectively the same.

With that said, I would argue this is a more interesting finding than the one I claimed, because it is more specific. The model is not casting around differently when unsure. It concentrates on the two signals that physically govern how much material enters the die and how hard it is compressed, which is what determines tablet hardness. It was given ten raw channels and four laboratory numbers and was never told which of them mattered.

The wording in Section 6.4 should have described amplification of specific channels, evidenced by the ratios, rather than reordering, evidenced by τ .

Question 9

The report states plainly that this study covers one production line and one press (Section 6.5). Yet the framing throughout (in the title, the abstract, and the stated contribution) is pitched at the level of trustworthy deep learning for pharmaceutical manufacturing in general. If a second site or press were tested tomorrow, which parts of MT-TrajNet's design do you expect would fail to transfer, and why was the framing not scoped more narrowly given this limitation was already known?

Answer 9. Let me take the framing part first, because I do not really have a defence for it. I knew about this limitation from the start. Section 6.5 states it plainly. The title and abstract should have carried the same scope and they do not. "On industrial compression trajectories" sounds like a general claim when the evidence behind it is one press and one dataset. A more honest title would have named the dataset outright or called the work a case study. I would change it now if I could.

On what would fail to transfer, I would separate three layers.

What I would expect to transfer is the method rather than the model. Grouping cross-validation by product code to prevent leakage; cross-fitting the calibration scale so no batch is calibrated on its own prediction; testing distribution shift against a matched in-regime control; stratifying attribution by the model's own confidence. None of these depend on the equipment. They are protocol, and they would apply to any instrumented press.

What would need refitting but not redesigning: the normalisation statistics, which are fold-specific in any case; the calibration scale factors, which would certainly differ on another press; and the 6,000-step sequence length, which was chosen for this dataset's run lengths and would need revisiting for a press with different batch durations.

What I would expect to fail is two things specifically.

The first is the learned encoder weights. The ten channels are specific to this press. Another press may carry different sensors, sample at a different rate, or measure the same nominal quantity differently. Channel ordering matters directly to my attribution result, so a different sensor set would invalidate the die fill and main compression finding until it was re-established from scratch.

The second, and the one I expect would fail hardest, is anything that depends on the hardness distribution. The $r = 0.941$ relationship says accuracy depends on how close a product sits to the centre of the training distribution. A press making a different product mix has a different centre, so the model would be accurate in a different place. Transfer would not simply be a matter of the model performing worse; it would be a matter of it being good over a different range.

That last point tells you what a useful transfer test would look like. Not whether accuracy holds up, since I would expect that to move. Whether the calibrated uncertainty still flags unfamiliar regimes on new equipment. That is the property the thesis actually claims, so that is the one to test first.

Question 10

Your multi-task model shares one encoder across four targets and trains with an unweighted sum of four evidential losses (Section 4.3). This is justified only by the targets being standardised to the same scale. But equal loss scale does not guarantee equal gradient contribution, especially once the evidential regularizer and the differing difficulty of each task come into play. How do you know that one of the four tasks (dissolution, for instance, which shows the only significant multi-task gain) is not simply dominating the shared representation rather than genuine positive transfer happening across all four targets?

Answer 10. You are right that equal loss scale does not mean equal gradient contribution. I never logged per-task gradient norms while training, so I cannot rule out an imbalance by measuring it. That is a real gap in the study.

What I can do is look at what came out, because the domination idea makes a prediction. If dissolution were taking over the shared encoder, the other three should be hurt by sharing, or at least left flat.

Target	Multi-task	Single-task	Improvement	p
Dissolution	3.261	3.547	8.1%	0.012
Hardness	7.698	8.310	7.4%	0.800
Tensile	0.345	0.364	5.2%	0.830
Weight RSD	0.585	0.595	1.7%	0.904

On the fold means, all four targets improve, and hardness gains 7.4 per cent, almost as much as dissolution’s 8.1 per cent.

I should not stop there, though, because the means are covering something up. Look at the per-fold differences instead of the averages and it is far less tidy:

Target	Per-fold difference (single – multi)	Multi-task better on
Dissolution	+0.310, +0.246, +0.301	3 of 3 folds
Hardness	+3.154, +0.086, −1.404	2 of 3 folds
Weight RSD	−0.002, +0.098, −0.066	1 of 3 folds
Tensile	−0.019, −0.039, +0.114	1 of 3 folds

Dissolution is the only one that improves everywhere. Weight RSD and tensile are actually worse under sharing on two folds each, and in both cases the mean-level gain rests on a single fold. Hardness gains on two and loses on the third.

So the honest version of my argument is narrower than the report implies. What I can show is that dissolution, the task you suspect of dominating, is not gaining at hardness’s expense. Hardness improves on two folds, including a big 3.154 gain on fold 1, which is not what domination would produce. What I cannot show is steady positive transfer to all four targets, and the report should not have suggested otherwise.

Dissolution reaches significance because its per-fold differences are consistent, all three between +0.246 and +0.310. The other three do not, because their differences change sign between folds, and a paired test on three folds cannot find an effect when the sign flips.

I want to correct something I could easily have got wrong here. It is tempting to blame this on MT-TrajNet being unstable across folds, but that is not what the data says. The multi-task hardness runs are stable, at 7.900, 7.428 and 7.767, a spread of 1.06 times. It is the single-task hardness runs that swing, at 11.054, 7.514 and 6.363, a spread of 1.74 times. The instability is in the baseline I am comparing against, not in my own model.

There is also a physical reason to expect genuine transfer rather than one task taking over. Hardness and tensile strength both depend on compaction, and dissolution depends on it as well. The four targets share upstream causes in the process, which is exactly the setting where a shared representation should help several tasks at once rather than only one.

So my position is that what I observed does not fit domination, because domination would mean the other tasks got worse, and all four got better on the means. But I accept this is indirect evidence about the effect, not direct evidence about the gradients themselves. The direct tests would be logging per-task gradient norms through the shared encoder during training, or comparing against a learned task-weighting scheme. I ran neither.

There is one more thing I should put on the record here, because it goes to the heart of what you are asking and I would rather raise it than have it found. My proposal specified the uncertainty weighting scheme of Kendall et al. (2018), which treats each task’s homoscedastic uncertainty as a learnable parameter and balances the tasks automatically. That is exactly the mechanism that would have answered your concern, because it removes the assumption that equal loss scale means equal gradient contribution. The model I built does not use it. It uses an unweighted sum instead.

I have gone back through the submitted report and it does not flag that change anywhere. Section 4.3 presents the equal sum as the design choice and argues that standardising the targets makes it sufficient, without noting that something different had been planned. That is a gap in the report. The change should have been stated and justified rather than passed over quietly, and I never tested whether the simpler choice cost anything.

Closing note

Several of these questions found things I had not checked, and two found genuine errors in the submitted report: the missing multiple-comparison correction raised in Question 1, and the misinterpretation of Kendall's τ raised in Question 8. I would rather set both out plainly than argue around them.

Where that leaves the study is worth setting out plainly. The out-of-distribution result is the part I am most confident about, because it was attacked from two directions and held: the uncertainty tracks a real change of operating regime rather than mere novelty, and that survives both the 24-code analysis and the within-code-13 check. The attribution finding is real as an observation, but it is about amplification of two specific channels rather than reordering, and it has no significance test behind it. The multi-task result is narrower than the report suggests. Dissolution improves on every fold, while weight uniformity and tensile strength are worse under sharing on two folds each, so I cannot claim consistent transfer across all four targets.

The weaknesses reported in the thesis remain weaknesses, and Question 3 has shown me there was a further one I should have investigated.

Thank you again for the care you have taken over this. The questions were harder than I expected, and the work is more honestly described now than it was when I submitted it.